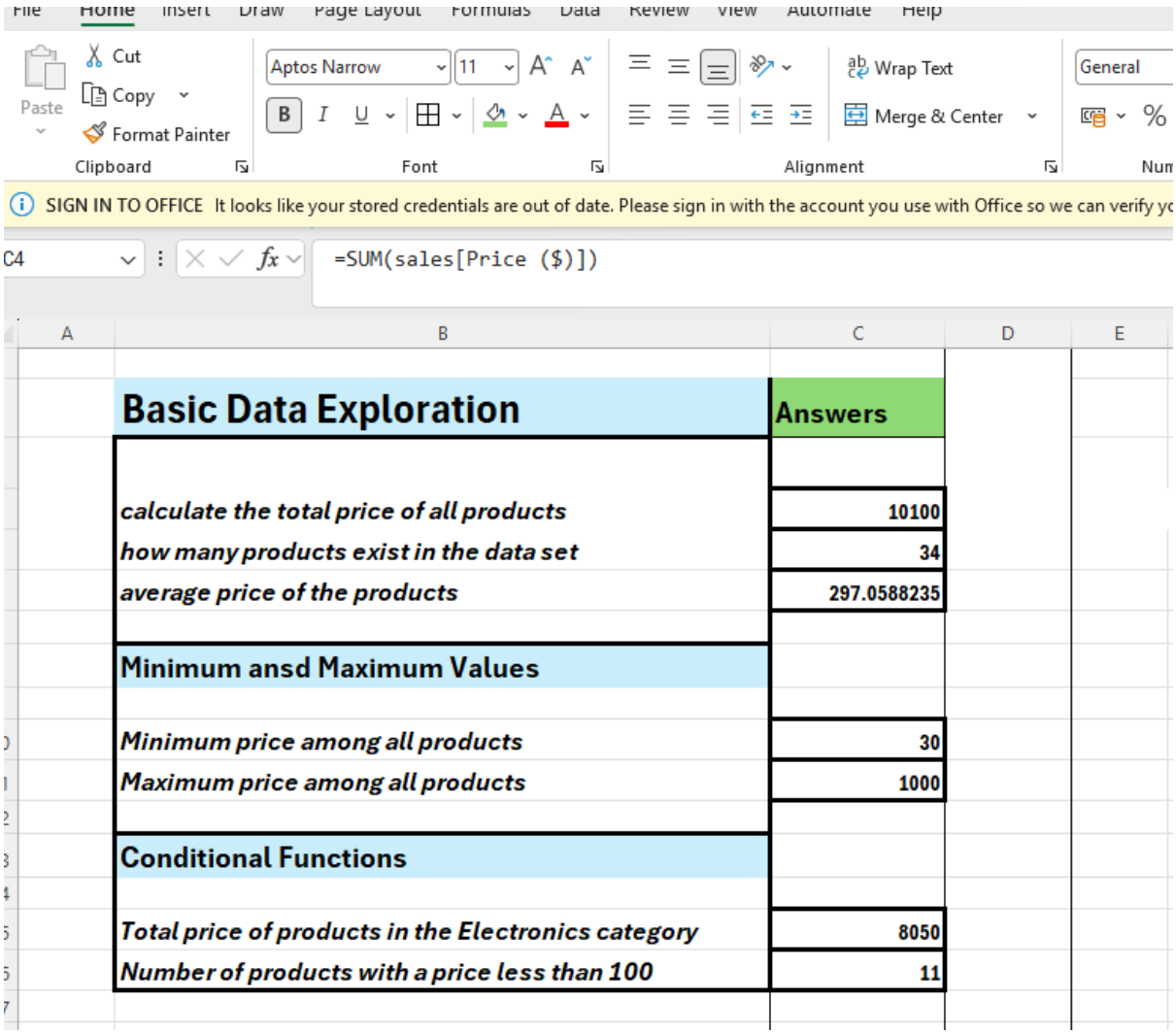
Basic Data Exploration

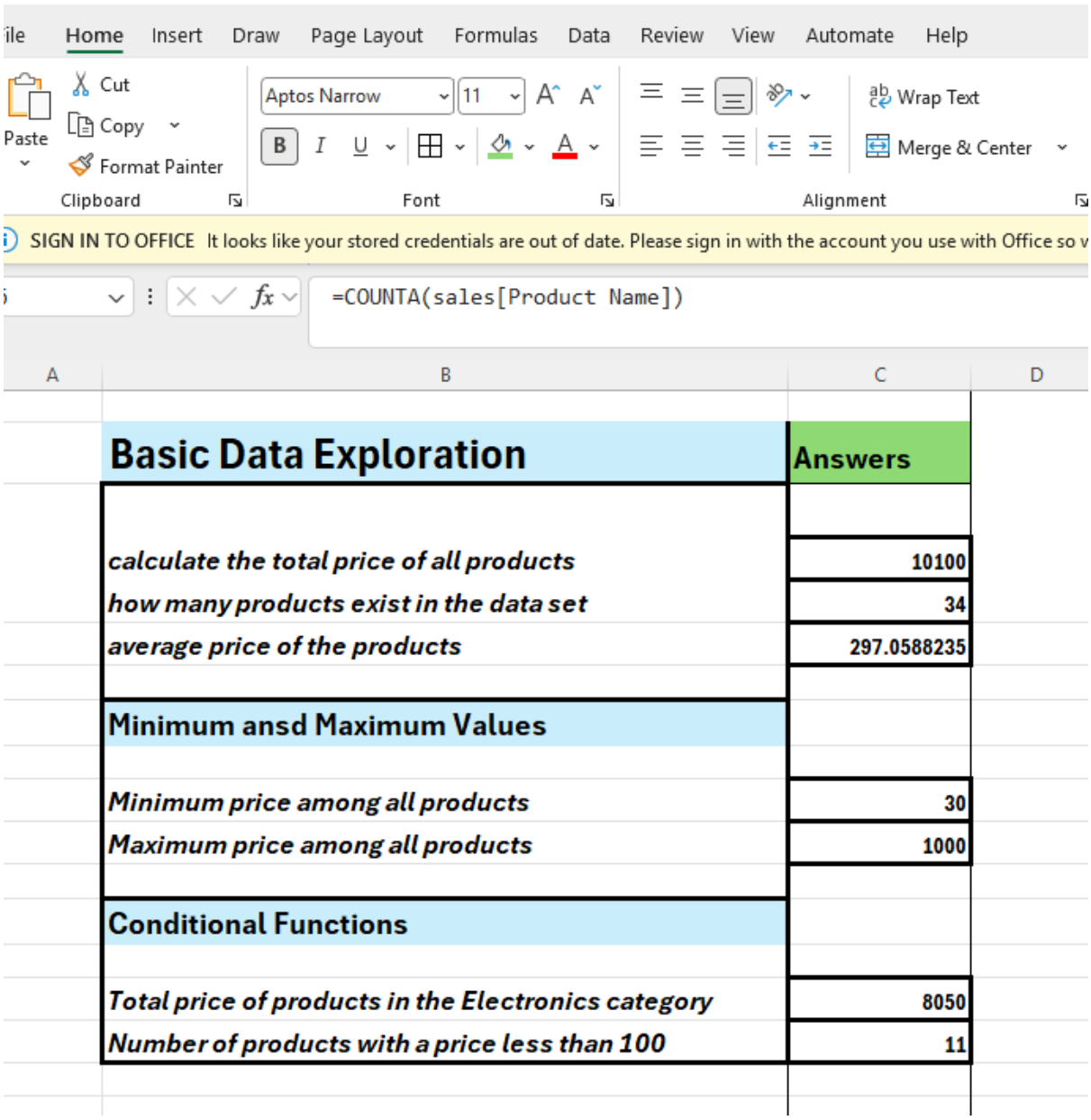
Assignment -1

Calculate the total price of all products in the dataset using SUM



Basic Data Exploration

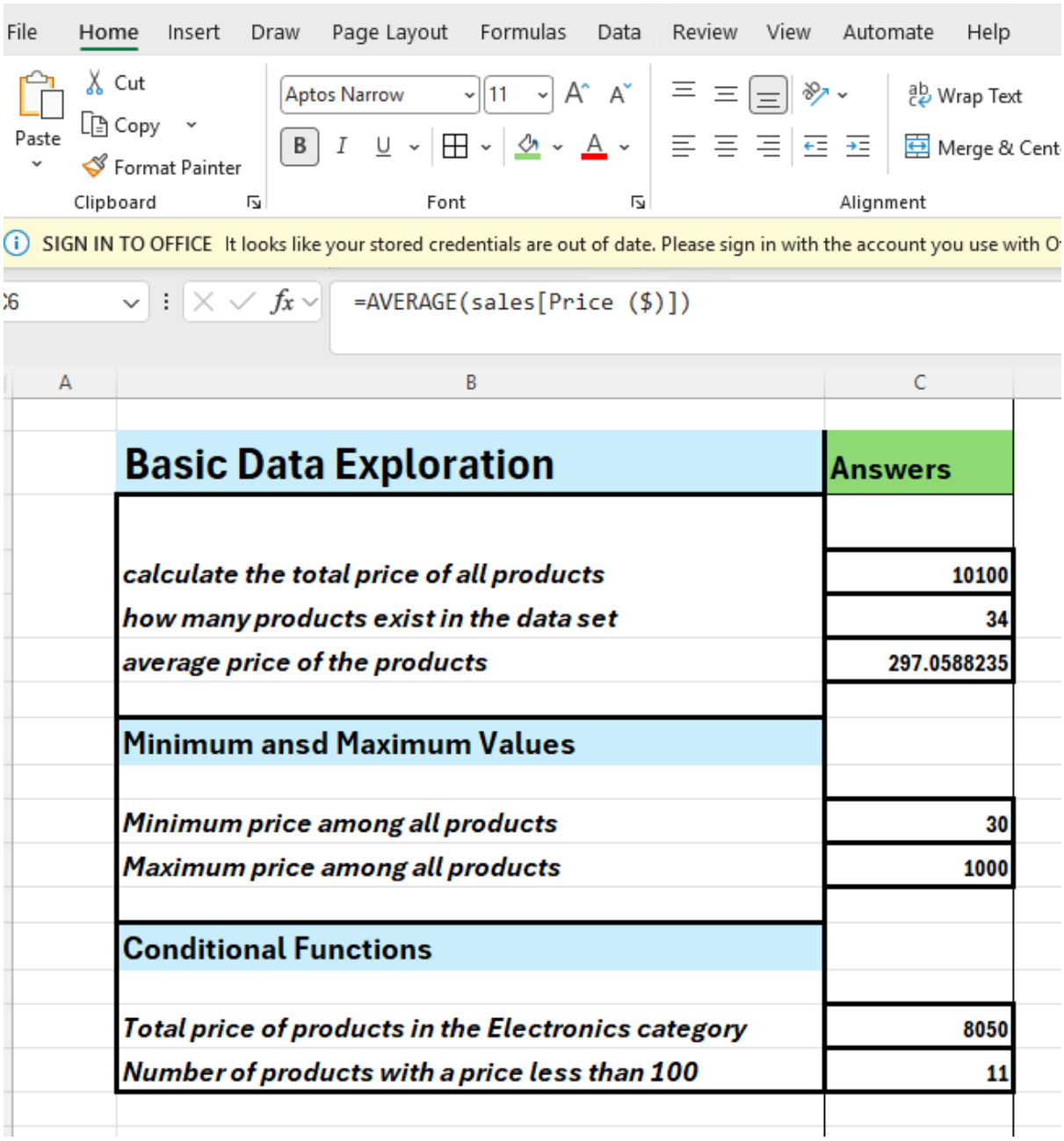
Determine how many products exist in the dataset using COUNTA



Basic Data Exploration	Answers
calculate the total price of all products	10100
how many products exist in the data set	34
average price of the products	297.0588235
Minimum ansd Maximum Values	
Minimum price among all products	30
Maximum price among all products	1000
Conditional Functions	
Total price of products in the Electronics category	8050
Number of products with a price less than 100	11

Basic Data Exploration

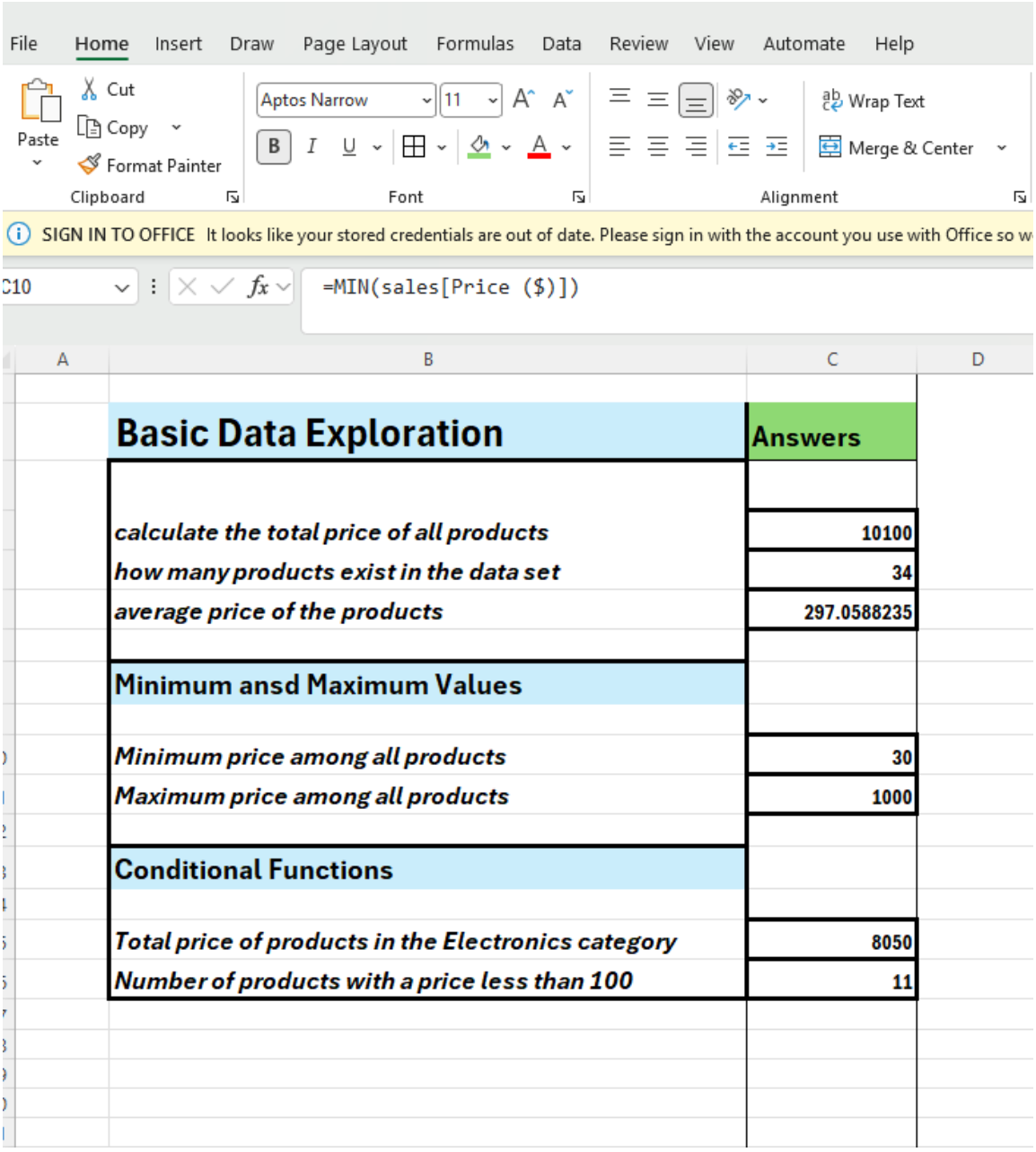
Calculate the average price of the products using AVERAGE



Basic Data Exploration

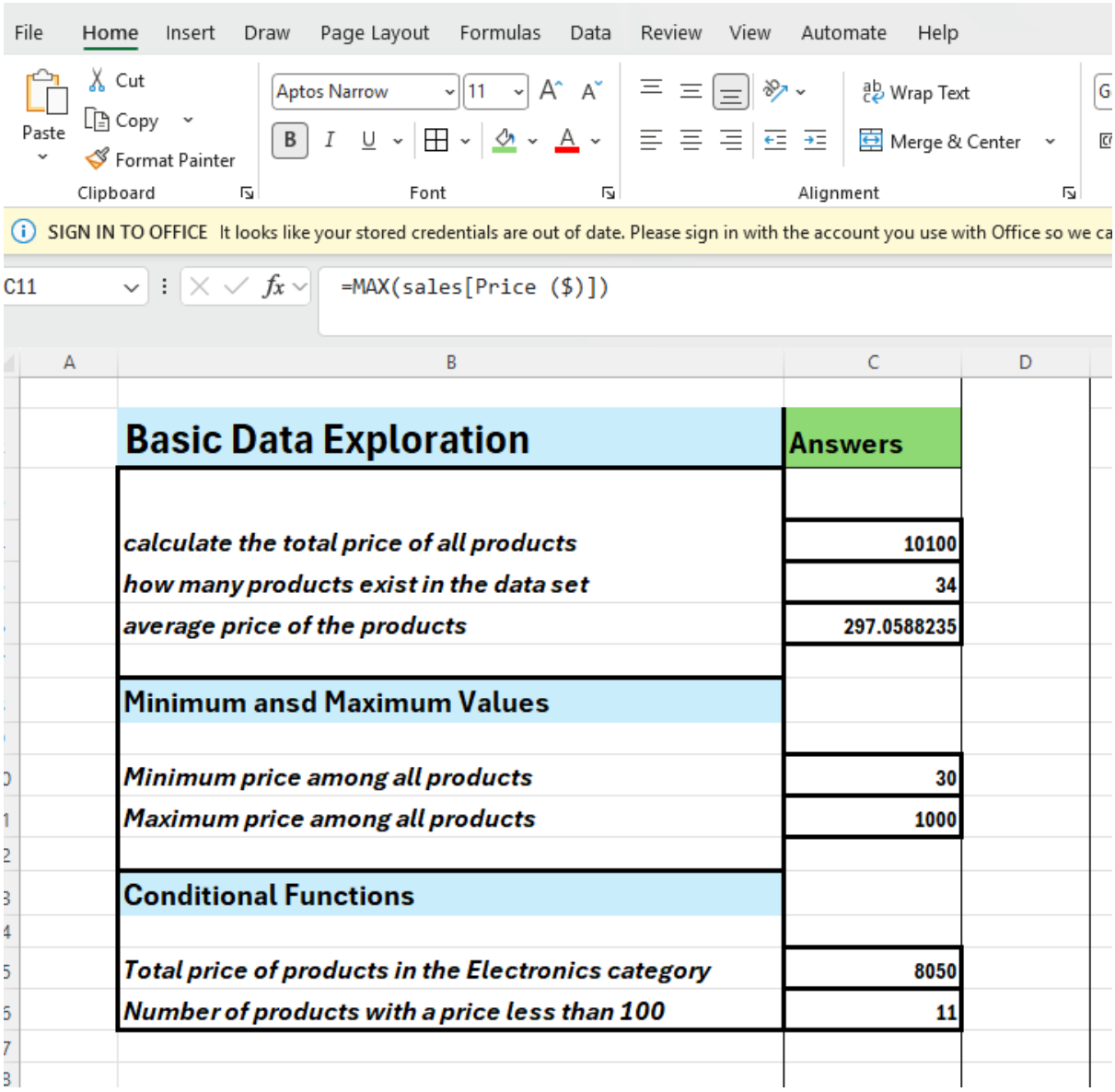
Minimum and Maximum values

1. Minimum price of products



Basic Data Exploration

2. Maximum price of products



Basic Data Exploration

Logical Functions

1. Create a new column named Price Range.

Use the IF function to categorize products based on their price:

- **If Price ≥ $500 → classify as High Price**
- **If Price < $500 → classify as Standard Price**

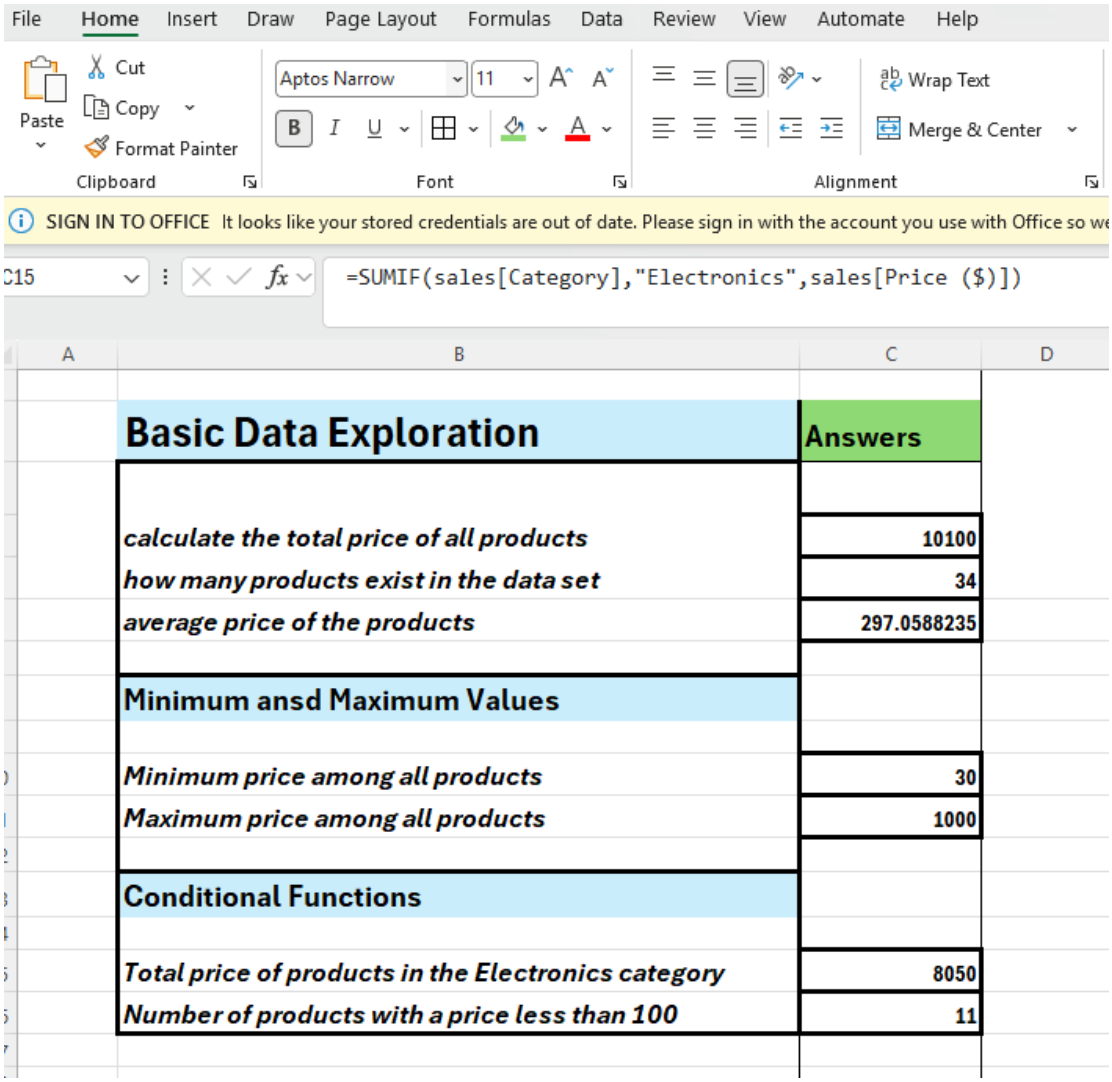
G2 `=IF([@[Price ($)]]>=500,"High Price","Standard Price")`

Product I	Product Nam	Brand Nam	Price ($	Quantit	Category	Price Range	Day	Country code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA	JUL
16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES	APR
07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA	MAR
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA	APR
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA	MAY
02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA	DEC
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN

Basic Data Exploration

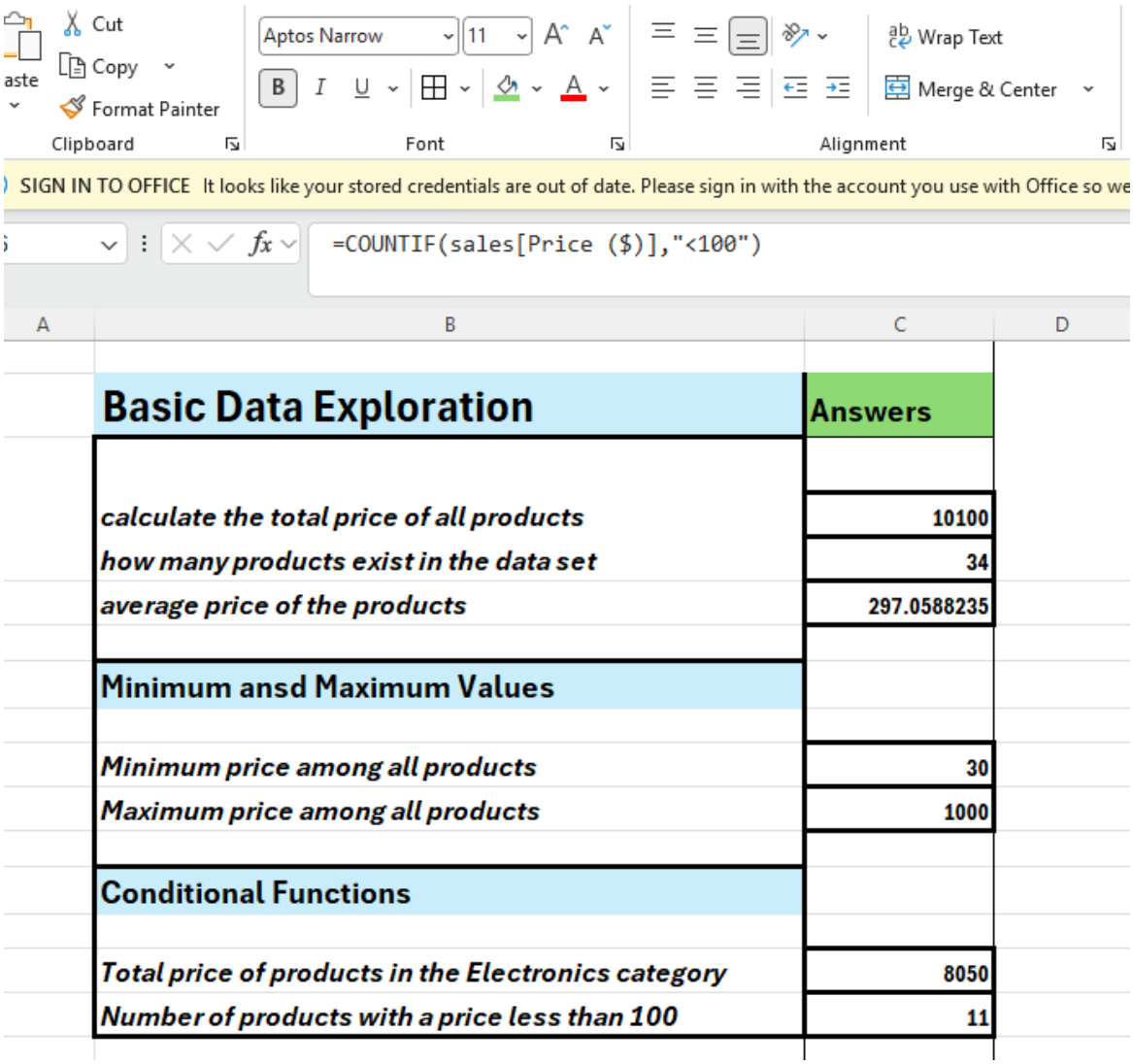
Conditional Functions

- **Use the SUMIF function to calculate the total price of products in the Electronics category.**



Basic Data Exploration

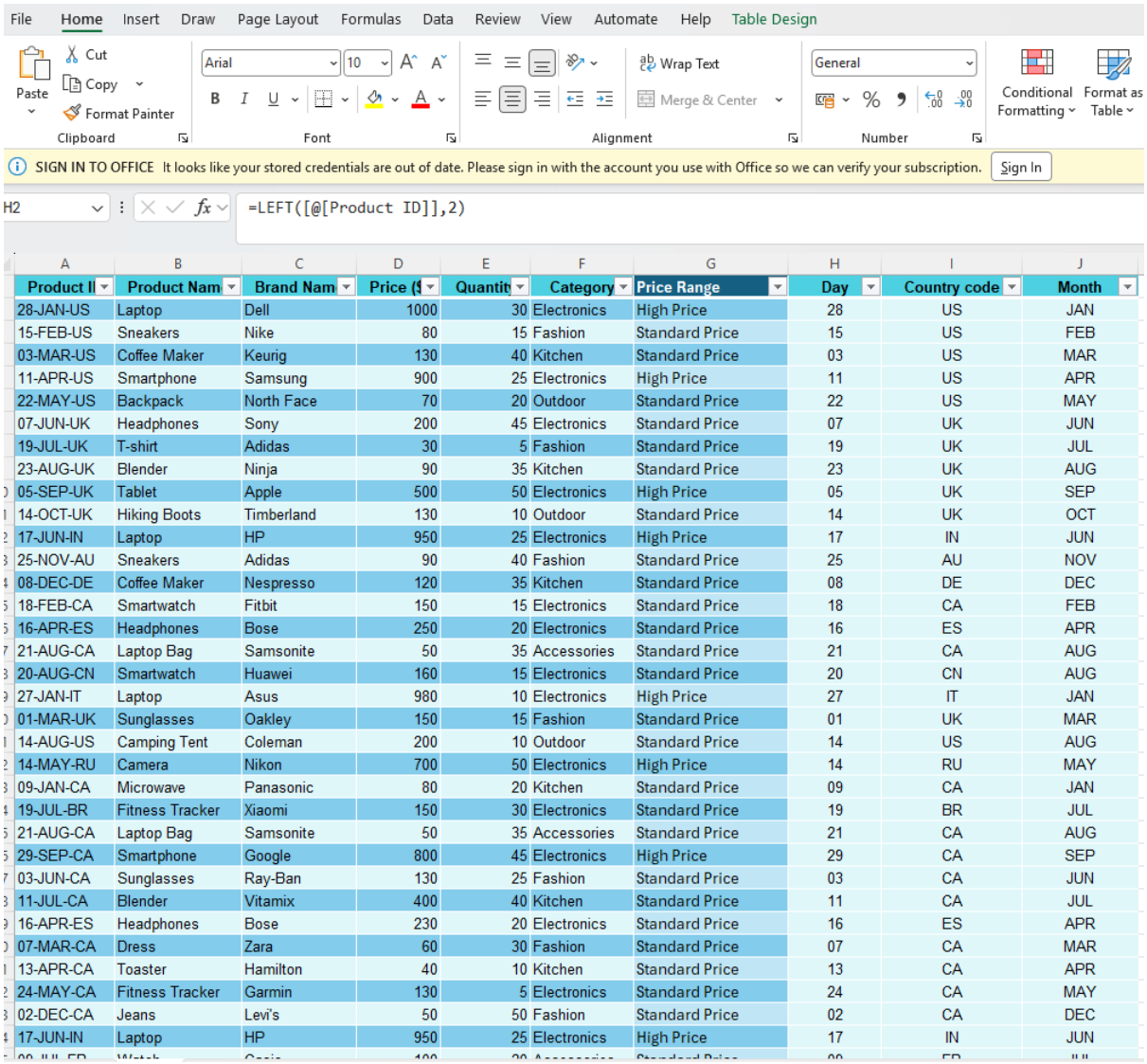
- **Use the COUNTIF function to determine the number of products with a price less than $100**



Basic Data Exploration

Text Formatting – LEFT, RIGHT, MID

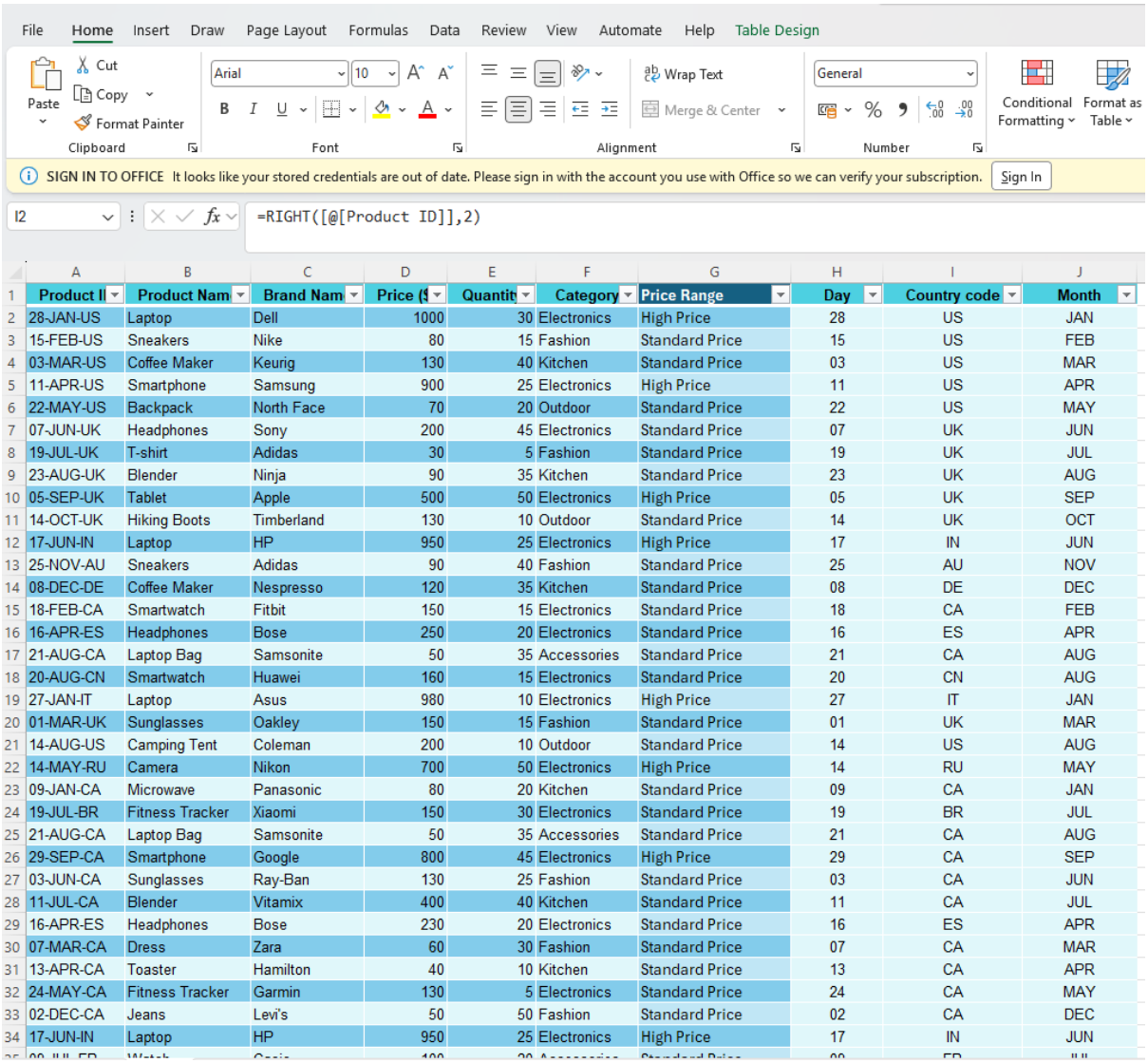
- **Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function**



Product Il	Product Nam	Brand Nam	Price ($	Quantit	Category	Price Range	Day	Country code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN
11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA	JUL
16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES	APR
07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA	MAR
13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA	APR
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA	MAY
02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA	DEC
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN

Basic Data Exploration

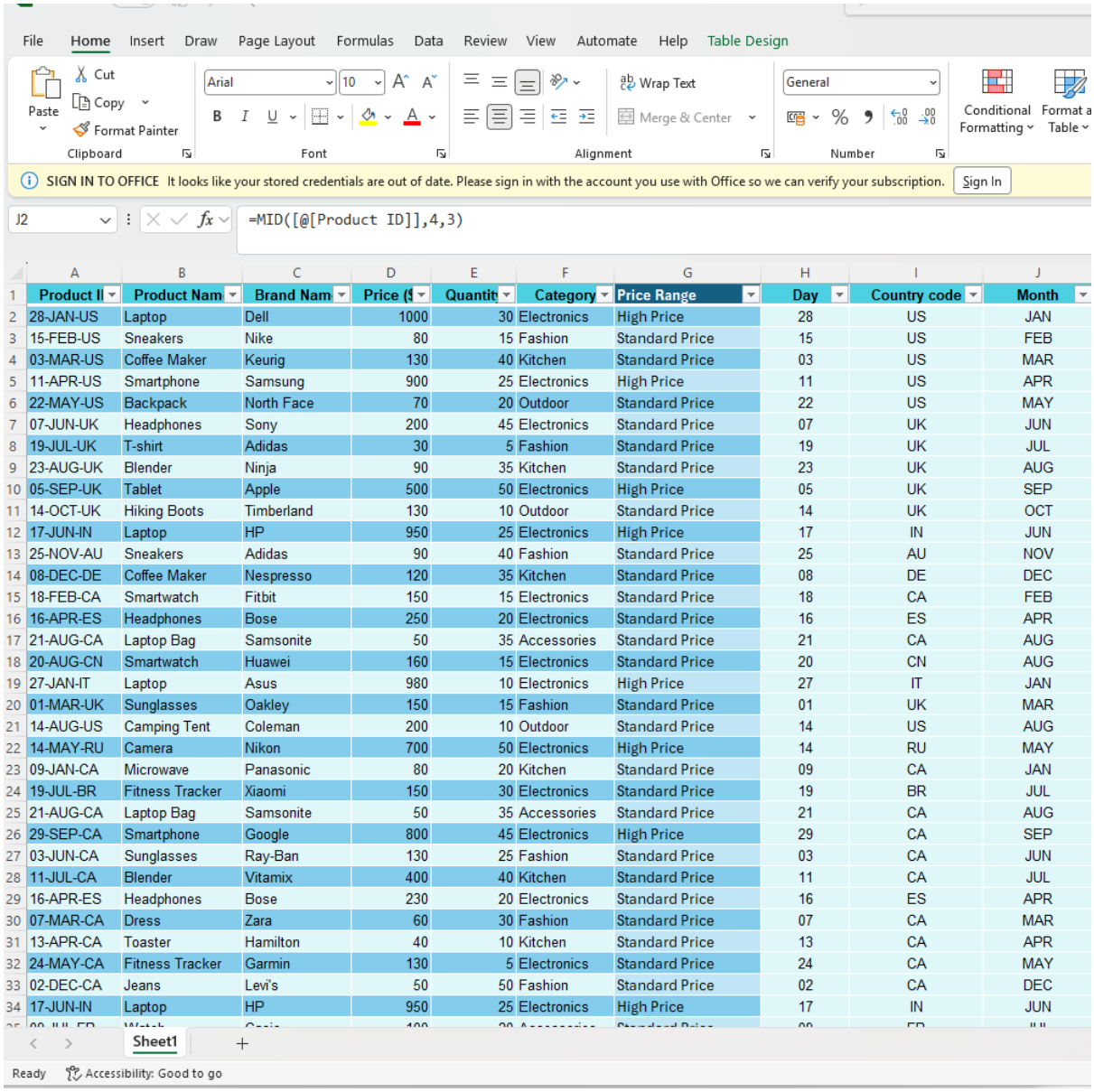
- **Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function**



	A	B	C	D	E	F	G	H	I	J
1	Product Il	Product Nam	Brand Nam	Price ($	Quantit	Category	Price Range	Day	Country code	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA	JUL
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES	APR
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA	MAR
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA	APR
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA	MAY
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA	DEC
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN

Basic Data Exploration

- **Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function**



	A	B	C	D	E	F	G	H	I	J
1	Product Il	Product Nam	Brand Nam	Price ($	Quantit	Category	Price Range	Day	Country code	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA	JUL
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES	APR
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA	MAR
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA	APR
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA	MAY
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA	DEC
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN